Traceable Entity Reconciliation for Enterprise Knowledge Graphs

Technical Supplement: Paired Query Controls, Frozen Policies, and WDC Split Audit

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October 13, 2026

This supplement documents the additional analyses integrated into the revised manuscript. Original E1–E5 matching, robustness, and scale outputs remain unchanged. No model, calibrator, retrieval setting, or acceptance threshold was refitted. Code and supporting research materials are available on GitHub: *[GitHub repository link to be added]*.
The parent manuscript's assistance disclosure applies to the implementation, checks, and drafting of this supplement. Numerical entries are generated from measured revision outputs. No participants were recruited, no human study was performed, and no fresh retraining is claimed.

1 Protocol and verification

The four graph tasks retain their original test views and common candidate pools. Original saved probabilities and method-specific validation thresholds define attribute-only (A) and gated-context (C) maps. Both methods process pairs in descending score order with lexicographic endpoint tie breaking, same-type eligibility, and one accepted partner per endpoint. Canonical identifiers are hashes of the endpoint pair. Gold labels are used only for evaluation. The same source projection, eight queries, and oracle apply to both methods.

There are four datasets, three seeds, two methods, and eight queries: 192 fresh method/query executions, representing 96 paired comparisons. All SQL and graph answer sets are equal, all oracle workloads are nonempty, and contextual canonical links and 96 contextual query metrics reproduce the original records. The 24 method-specific identity maps satisfy one-to-one constraints. All inspected original inputs retain their SHA-256 values. The additional control manifest records 748 passed checks. The full project test suite was rerun separately: 36 passed, zero failures, errors, or skips.

Query F1 is $2|E \cap A|/(|E| + |A|)$ for oracle set E and actual answer set A. Table values below are arithmetic means of seed-specific F1, unless explicitly labelled summed-trial totals. Seeds share original source populations; differences are descriptive. No new query bootstrap, pooled confidence interval, multiplicity adjustment, or confirmatory significance test is asserted.

2 Paired query comparison

Q1 returns reconciled pairs sharing projected tags; Q3 returns attributed amount/count; Q5 returns positive-value transaction links; Q6 returns cross-source identities; Q7 returns tag-grouped amount/count; Q8 returns transaction source-table lineage. Q2 and Q4 are source-local controls, equal to one for both methods. Amounts remain cent-rounded source units, and Q7 tag totals overlap.

Table 1: Paired three-seed means for the six identity-sensitive queries.

Dataset	Query	Attribute F1	Context F1	Δ F1
WWI	Q1	0.7457	0.7998	+0.0541
WWI	Q3	0.8147	0.8453	+0.0306
WWI	Q5	1.0000	0.9980	-0.0020
WWI	Q6	0.8021	0.8318	+0.0297
WWI	Q7	0.7742	0.8164	+0.0422
WWI	Q8	0.9187	0.9307	+0.0120
Olist	Q1	0.7170	0.7942	+0.0772
Olist	Q3	0.6942	0.7426	+0.0483
Olist	Q5	0.6815	0.7287	+0.0472
Olist	Q6	0.6942	0.7426	+0.0483
Olist	Q7	0.6928	0.7412	+0.0484
Olist	Q8	0.6815	0.7287	+0.0472
DataCo	Q1	0.7891	0.7386	-0.0504
DataCo	Q3	0.7897	0.7468	-0.0430
DataCo	Q5	0.7786	0.7424	-0.0362
DataCo	Q6	0.7897	0.7468	-0.0430
DataCo	Q7	0.7851	0.7447	-0.0404
DataCo	Q8	0.7786	0.7424	-0.0362
BPI2019	Q1	0.0667	0.5609	+0.4943
BPI2019	Q3	0.0630	0.5283	+0.4652
BPI2019	Q5	0.0600	0.6025	+0.5425
BPI2019	Q6	0.0630	0.5283	+0.4652
BPI2019	Q7	0.0635	0.5600	+0.4966
BPI2019	Q8	0.0635	0.5938	+0.5303

Olist and BPI improve on all six template means; DataCo declines on all six. WWI improves on five templates, with a small Q5 decline from 1.0000 to 0.9980. Query consequences depend on which transaction or aggregate tuples an identity decision changes. BPI compares detailed histories with coarse modal descriptors, not vendor names. Conditional pair F1, preconstraint end-to-end pair F1, canonical-link F1, and query-answer F1 have different denominators.

3 Canonical links after identical constraints

The full truth universe contains one A/B pair per held-out source group. Counts sum three seed trials over a shared population; they are not independent production decisions. Precision is true accepted links divided by accepted links; recall uses all truth pairs, including unretrieved pairs. This summed-trial F1 can differ from the arithmetic mean Q6 F1.

4 Absent-support policy diagnosis

For BPI, removing category-support edges sets every candidate's gate to zero in each seed. Frozen attribute scores are independently replayed and match the saved values within 10^{-12}. The contextual fallback scores equal those same probabilities exactly. The original policies remain different: a contextual acceptance threshold higher than the attribute threshold changes decisions for probabilities between the two thresholds. The table reproduces the original no-support preconstraint stress counts and end-to-end F1, whose truth

Table 2: Postconstraint canonical-link results using summed-trial denominators.

Dataset	Map	Truth	Accepted	False	Precision	Recall	F1
WWI	A	147	101	1	0.9901	0.6803	0.8065
WWI	C	147	107	1	0.9907	0.7211	0.8346
Olist	A	1749	1009	51	0.9495	0.5477	0.6947
Olist	C	1749	1092	37	0.9661	0.6032	0.7427
DataCo	A	900	692	63	0.9090	0.6989	0.7902
DataCo	C	900	636	62	0.9025	0.6378	0.7474
BPI2019	A	828	201	168	0.1642	0.0399	0.0641
BPI2019	C	828	372	54	0.8548	0.3841	0.5300

denominator is 276 per seed. Retained candidate positives are 205, 211, and 209. These are pair results before one-to-one canonicalization.

Table 3: Identical fallback probabilities evaluated at the original frozen thresholds.

Seed	Policy	$h = 0$ pairs	Threshold	TP	FP	End-to-end F1
17	A	7850	0.0530	34	199	0.1336
17	C	7850	0.3355	0	0	0.0000
29	A	7887	0.0788	18	165	0.0784
29	C	7887	0.2594	3	9	0.0208
43	A	8049	0.0440	62	717	0.1175
43	C	8049	0.4019	0	0	0.0000

The mean end-to-end F1 values are 0.1099 for A and 0.0069 for C. Evidence fallback does not guarantee decision fallback. No aligned fallback policy was fitted or tested in this revision. Such a policy would require development-only selection followed by a fresh evaluation; using test results to choose it would invalidate the comparison.

5 Independent WDC split audit

The benchmark paper describes offer-disjoint train, validation, and test splits [1]. The retained pairwise archives used here show a discrepancy. The independent parser reads the publisher's `id_left` and `id_right` as string offer IDs, unions both endpoints in each split, and intersects these offer sets. It does not substitute cluster IDs, generated source UIDs, or pair IDs. Every intersecting ID has identical cluster, title, description, brand, price, and currency fields across the relevant files. Hashed example records are retained without reproducing offer text.

All 27 independently computed cells reproduce the original saved split audit. Eighteen have training/test and/or validation/test intersections; all nine 100%-unseen cells have none. The diagnostic subset retains only test pairs with neither endpoint in training or validation. No official pair list, training data, threshold, or reported matching output was modified. The reason for the discrepancy with the intended split design remains unresolved; these findings concern the exact archived bytes below and do not characterize every benchmark release.

Table 4: Offer intersections across all 27 cells. Each full test contains 4,500 pairs.

Hardness	Size	Unseen	Train/test	Val./test	Disjoint test pairs
20	small	0	0	4	4469
20	small	50	97	105	2918
20	small	100	0	0	4500
20	medium	0	0	4	4469
20	medium	50	155	105	2510
20	medium	100	0	0	4500
20	large	0	0	4	4469
20	large	50	269	105	1889
20	large	100	0	0	4500
50	small	0	0	4	4465
50	small	50	110	106	2836
50	small	100	0	0	4500
50	medium	0	0	4	4465
50	medium	50	162	106	2498
50	medium	100	0	0	4500
50	large	0	0	4	4465
50	large	50	274	106	1946
50	large	100	0	0	4500
80	small	0	0	4	4476
80	small	50	110	103	2853
80	small	100	0	0	4500
80	medium	0	0	4	4476
80	medium	50	159	103	2564
80	medium	100	0	0	4500
80	large	0	0	4	4476
80	large	50	272	103	1998
80	large	100	0	0	4500

5.1 Archive identity and exact members

Archives were retained from the October 2026 study acquisition, not downloaded anew for this revision. Full archive metadata is in `data/wdc_archives.json`. Exact member names and decompressed SHA-256 values for every cell are in `data/wdc_split_audit.csv`. Archive URLs use the publisher directory shown below, with names `20pair.zip`, `50pair.zip`, and `80pair.zip`.
`https://data.dws.informatik.uni-mannheim.de/largescaleproductcorpus/data/wdc-products/`

Table 5: Exact identities of the three audited publisher archives.

Retained archive	Bytes	SHA-256 (two consecutive lines)
wdc-products-20pair.zip	20,500,888	`7d23e706ed56d5e4ae53088251240725` `492a3b1e7425e3cee379c0edb1044d25`
wdc-products-50pair.zip	22,112,083	`57fca0c66bd1d7a09298dfb5df4ed642` `8251487517e9cfbce209aad99388359d`
wdc-products-80pair.zip	21,985,075	`b2044939cee5ea6f12148a2f3551508d` `e3cb77660dfc91767c44daaf9d8a9c4a`

For hardness $H \in \{20, 50, 80\}$, the exact member prefix is `wdcproducts`H`cc`$(100-H)$`rnd`. Training and

validation append `000un_train_S.json.gz` and `000un_valid_S.json.gz`, where S is small, medium, or large. Tests append `000un_gs.json.gz`, `050un_gs.json.gz`, or `100un_gs.json.gz`. For example:
`wdcproducts20cc80rnd000un_train_small.json.gz`
`wdcproducts20cc80rnd000un_valid_small.json.gz`
`wdcproducts20cc80rnd050un_gs.json.gz`

This example has 97 training/test and 105 validation/test overlapping offers and retains 2,918 offer-disjoint test pairs. Its 0%-unseen test has zero and four overlapping offers and retains 4,469 pairs. Fully unseen tests retain all 4,500. Overlap examples and attribute/cluster digests are in `data/wdc_overlap_examples.json`.

6 Per-seed query denominators and differences

The following table retains all 96 paired query cells, including local controls. $|E|$ denotes oracle answers; $|A|$ and $|C|$ denote actual answer counts. Full true-positive counts, precision, and recall are in `data/query_controls.csv`. Each query mean above is the arithmetic mean of its three rows here.

Table 6: Full paired per-seed query records.

Dataset	Seed	Q	$\|E\|$	$\|A\|$	$\|C\|$	A F1	C F1	Δ
WWI	17	Q1	24	10	13	0.5882	0.7027	+0.1145
WWI	17	Q2	9446	9446	9446	1.0000	1.0000	+0.0000
WWI	17	Q3	47	27	30	0.7297	0.7792	+0.0495
WWI	17	Q4	5047	5047	5047	1.0000	1.0000	+0.0000
WWI	17	Q5	1067	1067	1054	1.0000	0.9939	-0.0061
WWI	17	Q6	49	28	31	0.7273	0.7750	+0.0477
WWI	17	Q7	37	18	22	0.6545	0.7458	+0.0912
WWI	17	Q8	2298	1834	1909	0.8877	0.9075	+0.0198
WWI	29	Q1	30	24	24	0.8889	0.8889	+0.0000
WWI	29	Q2	9446	9446	9446	1.0000	1.0000	+0.0000
WWI	29	Q3	47	40	41	0.9195	0.9318	+0.0123
WWI	29	Q4	8973	8973	8973	1.0000	1.0000	+0.0000
WWI	29	Q5	1067	1067	1067	1.0000	1.0000	+0.0000
WWI	29	Q6	49	41	42	0.8889	0.9011	+0.0122
WWI	29	Q7	40	34	34	0.8919	0.8919	+0.0000
WWI	29	Q8	2298	2155	2177	0.9589	0.9640	+0.0051
WWI	43	Q1	31	19	21	0.7600	0.8077	+0.0477
WWI	43	Q2	9446	9446	9446	1.0000	1.0000	+0.0000
WWI	43	Q3	47	31	33	0.7949	0.8250	+0.0301
WWI	43	Q4	6660	6660	6660	1.0000	1.0000	+0.0000
WWI	43	Q5	1067	1067	1067	1.0000	1.0000	+0.0000
WWI	43	Q6	49	32	34	0.7901	0.8193	+0.0292
WWI	43	Q7	41	26	28	0.7761	0.8116	+0.0355
WWI	43	Q8	2298	1917	1960	0.9096	0.9206	+0.0110
Olist	17	Q1	669	389	454	0.7221	0.7747	+0.0526
Olist	17	Q2	1408	1408	1408	1.0000	1.0000	+0.0000
Olist	17	Q3	583	325	355	0.6872	0.7186	+0.0313
Olist	17	Q4	2816	2816	2816	1.0000	1.0000	+0.0000
Olist	17	Q5	710	375	410	0.6581	0.6911	+0.0330
Olist	17	Q6	583	325	355	0.6872	0.7186	+0.0313
Olist	17	Q7	1179	654	714	0.6852	0.7163	+0.0311
Olist	17	Q8	710	375	410	0.6581	0.6911	+0.0330

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Dataset	Seed	Q	$\|E\|$	$\|A\|$	$\|C\|$	A F1	C F1	Δ
Olist	29	Q1	669	416	466	0.7410	0.8053	+0.0643
Olist	29	Q2	1408	1408	1408	1.0000	1.0000	+0.0000
Olist	29	Q3	583	363	374	0.7252	0.7628	+0.0376
Olist	29	Q4	2816	2816	2816	1.0000	1.0000	+0.0000
Olist	29	Q5	710	441	451	0.7246	0.7545	+0.0299
Olist	29	Q6	583	363	374	0.7252	0.7628	+0.0376
Olist	29	Q7	1179	732	755	0.7242	0.7622	+0.0379
Olist	29	Q8	710	441	451	0.7246	0.7545	+0.0299
Olist	43	Q1	669	369	465	0.6879	0.8025	+0.1146
Olist	43	Q2	1408	1408	1408	1.0000	1.0000	+0.0000
Olist	43	Q3	583	321	363	0.6704	0.7463	+0.0759
Olist	43	Q4	2816	2816	2816	1.0000	1.0000	+0.0000
Olist	43	Q5	710	390	435	0.6618	0.7406	+0.0788
Olist	43	Q6	583	321	363	0.6704	0.7463	+0.0759
Olist	43	Q7	1179	648	732	0.6689	0.7452	+0.0763
Olist	43	Q8	710	390	435	0.6618	0.7406	+0.0788
DataCo	17	Q1	1393	1142	978	0.8276	0.7465	-0.0811
DataCo	17	Q2	4558	4558	4558	1.0000	1.0000	+0.0000
DataCo	17	Q3	300	247	219	0.8190	0.7553	-0.0637
DataCo	17	Q4	9116	9116	9116	1.0000	1.0000	+0.0000
DataCo	17	Q5	2375	1969	1737	0.8094	0.7578	-0.0516
DataCo	17	Q6	300	247	219	0.8190	0.7553	-0.0637
DataCo	17	Q7	2529	2085	1818	0.8149	0.7550	-0.0599
DataCo	17	Q8	2375	1969	1737	0.8094	0.7578	-0.0516
DataCo	29	Q1	1393	1070	1048	0.7739	0.7751	+0.0012
DataCo	29	Q2	4558	4558	4558	1.0000	1.0000	+0.0000
DataCo	29	Q3	300	230	225	0.7774	0.7695	-0.0078
DataCo	29	Q4	9116	9116	9116	1.0000	1.0000	+0.0000
DataCo	29	Q5	2375	1833	1794	0.7586	0.7719	+0.0133
DataCo	29	Q6	300	230	225	0.7774	0.7695	-0.0078
DataCo	29	Q7	2529	1958	1925	0.7742	0.7827	+0.0084
DataCo	29	Q8	2375	1833	1794	0.7586	0.7719	+0.0133
DataCo	43	Q1	1393	989	851	0.7657	0.6943	-0.0714
DataCo	43	Q2	4558	4558	4558	1.0000	1.0000	+0.0000
DataCo	43	Q3	300	215	192	0.7728	0.7154	-0.0574
DataCo	43	Q4	9116	9116	9116	1.0000	1.0000	+0.0000
DataCo	43	Q5	2375	1698	1490	0.7680	0.6975	-0.0704
DataCo	43	Q6	300	215	192	0.7728	0.7154	-0.0574
DataCo	43	Q7	2529	1781	1566	0.7661	0.6965	-0.0697
DataCo	43	Q8	2375	1698	1490	0.7680	0.6975	-0.0704
BPI2019	17	Q1	1019	147	518	0.0806	0.5830	+0.5023
BPI2019	17	Q2	8588	8588	8588	1.0000	1.0000	+0.0000
BPI2019	17	Q3	276	52	135	0.0854	0.5499	+0.4645
BPI2019	17	Q4	27366	27366	27366	1.0000	1.0000	+0.0000
BPI2019	17	Q5	4308	638	2614	0.0708	0.6209	+0.5502
BPI2019	17	Q6	276	52	135	0.0854	0.5499	+0.4645
BPI2019	17	Q7	1082	206	550	0.0807	0.5821	+0.5014
BPI2019	17	Q8	4574	775	2709	0.0778	0.6143	+0.5365
BPI2019	29	Q1	1019	90	506	0.0325	0.5915	+0.5590
BPI2019	29	Q2	8588	8588	8588	1.0000	1.0000	+0.0000
BPI2019	29	Q3	276	32	135	0.0325	0.5693	+0.5369

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Dataset	Seed	Q	$\|E\|$	$\|A\|$	$\|C\|$	A F1	C F1	Δ
BPI2019	29	Q4	27366	27366	27366	1.0000	1.0000	+0.0000
BPI2019	29	Q5	4308	421	2491	0.0512	0.6298	+0.5786
BPI2019	29	Q6	276	32	135	0.0325	0.5693	+0.5369
BPI2019	29	Q7	1082	131	532	0.0297	0.5911	+0.5614
BPI2019	29	Q8	4574	438	2532	0.0483	0.6136	+0.5653
BPI2019	43	Q1	1019	316	409	0.0869	0.5084	+0.4215
BPI2019	43	Q2	8588	8588	8588	1.0000	1.0000	+0.0000
BPI2019	43	Q3	276	117	102	0.0712	0.4656	+0.3944
BPI2019	43	Q4	27366	27366	27366	1.0000	1.0000	+0.0000
BPI2019	43	Q5	4308	1606	2128	0.0582	0.5569	+0.4987
BPI2019	43	Q6	276	117	102	0.0712	0.4656	+0.3944
BPI2019	43	Q7	1082	494	433	0.0799	0.5069	+0.4270
BPI2019	43	Q8	4574	1751	2226	0.0645	0.5535	+0.4890

7 Reproduction and material index

Build the PDFs with `python build.py`; rebuilding the manuscripts does not execute experiments. The source ZIP supplies complete editable LaTeX, fixed figure sources, measured CSV/JSON outputs, and verification reports. In the complete KG-ERP project, use the pinned full-study environment and run the control script against an empty output directory:

```
python analysis/run_controls.py --project KG --output NEW
```

Replace `KG` with the complete project path and `NEW` with a new empty result directory. The command uses `results/all-datasets-v1`, the documented source views, and retained WDC archives. It writes only the new output directory; generic truth persistence is disabled. It loads BPI fitted estimators solely for frozen attribute-score replay, and never trains a model. Paths can point to a restored GitHub release using its documented materialization/acquisition workflow.

- `data/query_controls.csv`: 192 method/query rows with exact denominators and SQL/graph parity.
- `data/query_paired_differences.csv` and `data/query_paired_means.csv`: 96 seed-paired differences and 32 query means.
- `data/identity_controls.csv` and `data/identity_totals.csv`: 24 method/seed maps and eight summed-trial summaries.
- `data/gate_policy_diagnostic.csv`: six frozen BPI policy comparisons under absent support.
- `data/wdc_split_audit.csv`, `data/wdc_archives.json`, and `data/wdc_overlap_examples.json`: 27-cell audit, exact archive/member identities, and hashed examples.
- `data/verification.json`: 748 passed checks and input/output SHA-256 values; `data/tests.xml`: the independent 36-test run.

Reference

[1] R. Peeters, R. C. Der, and C. Bizer, "WDC Products: A Multi-Dimensional Entity Matching Benchmark," *Proc. EDBT*, pp. 22–33, 2024. DOI: 10.48786/edbt.2024.03.